Microeconomic Analysis

Static Optimisation

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Static Optimisation

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III. Optimisation with Inequality Constraints

Maximisation with Inequality Constraints

Suppose $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^m$ is a C^2 function. Consider the problem:

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

or equivalently

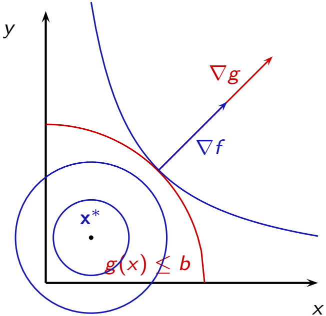
$$\max f(x_1, \ldots, x_n) \quad \text{s.t.}$$

$$\begin{cases} g_1(x_1, \ldots, x_n) \leq b_1 \\ \ldots \\ g_m(x_1, \ldots, x_n) \leq b_m \end{cases}$$

Geometric Intuition

The geometric intuition is similar to that with equality constraints except that now the gradient vector ∇f must point *outside* the region $g(\mathbf{x}) \leq \mathbf{b}$.

So if maximising $f(x, y)$ subject to $g(x, y) \leq b$ one has a picture such as:



Hence if a constraint is binding at $\mathbf{x}^*$, one must have $\nabla f = \lambda \nabla g$ with $\lambda \geq 0$.

However, we can also have an interior optimum $\mathbf{x}^*$, in which case $\nabla f = 0$.

Kuhn Tucker's Theorem

We define the Lagrangian L as before:

$$L(\mathbf{x}; \boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j (g_j(\mathbf{x}) - b_j)$$

And we define accordingly $D_x L(\mathbf{x}; \boldsymbol{\lambda})$ and $D_x^2 L(\mathbf{x}; \boldsymbol{\lambda})$

Kuhn-Tucker Theorem

If $\mathbf{x}^*$ is a local maximum of f subject to the constraints $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$, and the constraint qualification is satisfied at $\mathbf{x}^*$. Then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*)$ such that

$$D_x L(\mathbf{x}^*; \boldsymbol{\lambda}^*) = 0$$

$$\lambda_j^* \geq 0 \quad \text{and} \quad \lambda_j^*(g_j(\mathbf{x}^*) - b_j) = 0 \quad \text{for } j = 1, \ldots, m$$

The constraint qualification for $\mathbf{x}^*$ is as follows:

Let $\mathbf{g}_E$ be all the constraints which bind at $\mathbf{x}^*$, and e be the number of binding constraints. The constraint qualification is satisfied if the matrix $D\mathbf{g}_E(\mathbf{x}^*)$ is of full rank e.

Complementary Slackness Conditions

The conditions

$$\lambda_j^* \geq 0 \quad \text{and} \quad \lambda_j^*(g_j(\mathbf{x}^*) - b_j) = 0 \quad \text{for } j = 1, \ldots, m$$

are called the the complementary slackness conditions

1. λ_j for all j is non-negative as an agent only needs to be penalised if he wishes $g_j(x)$ to exceed b_j.
2. Note that if $g_j(\mathbf{x}^*) < b_j$, then $\lambda_j = 0$.

   That is λ_j is only positive if the jth constraint is satisfied with equality (and so it is possible the agent may wish to violate it).

In particular, these FOCs would capture:

- Interior optima: $\lambda_j^* = 0$ for all $j = 1, ..., m$, so $D_x L(\mathbf{x}^*; \boldsymbol{\lambda}^*) = Df(\mathbf{x}^*) = 0$.
- Constrained optima: $\lambda_j^* \geq 0$, $D_x L(\mathbf{x}^*; \boldsymbol{\lambda}^*) = 0$.

First Order Conditions

The Kuhn-Tucker Theorem means that (provided the constraint qualification is satisfied) you can look for a solution to the optimisation problem by looking for a solution to the following $n + m$ First Order Conditions:

$$\frac{\partial L}{\partial x_i}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = 0 \quad \text{for } i = 1, \ldots, n$$
$$\lambda_j \geq 0,\ g_j(\mathbf{x}^*) \leq b_j \quad \text{and} \quad \lambda_j(g_j(\mathbf{x}^*) - b_j) = 0 \quad \text{for } j = 1, \ldots, m$$

Interpretation of the Kuhn-Tucker multipliers:

As before, λ_j^* measures the sensitivity of the value of the objective function at $\mathbf{x}^*$ to a relaxation of the constraint $g_j(\mathbf{x}^*) \leq b_j$. If the jth constraint doesn't bind, then λ_j^* will be zero.

Second Order Conditions

Second Order Sufficient Conditions

Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfies the constraint qualification and the first order conditions.

1. If some of the constraints, $\mathbf{g}_E$, bind at $\mathbf{x}^*$ and $\mathbf{v}^T D_x^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*)\mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = 0$ then $\mathbf{x}^*$ is a strict local maximum.
2. If no constraints bind at $\mathbf{x}^*$ and $D^2 f(\mathbf{x}^*)$ is negative definite, then $\mathbf{x}^*$ is a strict local maximum.

For the first case, construct the bordered Hessian:

$$H = \begin{pmatrix} \mathbf{0} & D\mathbf{g}_E(\mathbf{x}^*) \\ D\mathbf{g}_E(\mathbf{x}^*)^T & D_x^2 L(\mathbf{x}^*; \boldsymbol{\lambda}^*) \end{pmatrix}$$

and check if the sign of the last $n - e$ leading principal minors alternate, ending with $(-1)^n$

Minimisation Problems with inequality Constraints

For minimisation problems:

$$\text{minimise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}^*) \geq \mathbf{b}$$

Define the Lagrangian as before

$$L(\mathbf{x}; \boldsymbol{\lambda}) = f(x) - \sum_{j=1}^{m} \lambda_j (g_j(\mathbf{x}) - b_j)$$

Then the Kuhn Tucker Theorem holds exactly as before and the FOCs and constraint qualifications are identical.

The sufficient second order conditions for a constrained strict local minimum is

$$\mathbf{v}^T D_x^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*)\mathbf{v} > 0 \text{ for all } \mathbf{v} \neq \mathbf{0} \text{ in } \mathbb{R}^n \text{ for which } D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = 0$$

or equivalently, the last $n - e$ leading principal minors all have sign of $(-1)^e$.

Optimisation Problems with Positivity Constraints

Consider the problem:

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}^*) \leq \mathbf{b},\ \mathbf{x} \geq \mathbf{0}$$

The second set of constraints are called positivity constraints.

These can be treated as normal inequality constraints, but Kuhn &Tucker proposed a reformulation of the Lagrangian to deal specifically with these positivity constraints.

Consider the Lagrangian as before

$$L(\mathbf{x}; \boldsymbol{\lambda}; \boldsymbol{\mu}) = f(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j (g_j(\mathbf{x}) - b_j) + \sum_{i=1}^{n} \mu_i x_i$$

Note that the CQ on the new constraints never fail (linear constraints never fail the CQ...)

Optimisation Problems with Positivity Constraints

Construct the FOC using Kuhn-Tucker Theorem:

$$\frac{\partial L}{\partial x_i}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \frac{\partial f}{\partial x_i}(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{x}) + \mu_i = 0 \quad \text{for } i = 1, \ldots, n \quad (1)$$

$$\lambda_j \geq 0,\ g_j(\mathbf{x}^*) \leq b_j \quad \text{and} \quad \lambda_j(g_j(\mathbf{x}^*) - b_j) = 0 \quad \text{for } j = 1, \ldots, m \quad (2)$$

$$\mu_i \geq 0,\ x_i \geq 0 \quad \text{and} \quad \mu_i x_i = 0 \quad \text{for } i = 1, \ldots, n \quad (3)$$

We can rearrange (1) and (3) so that we get rid of μ_i:

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{x}) \leq 0 \quad \text{for } i = 1, \ldots, n \quad (1)$$

$$\lambda_j \geq 0,\ g_j(\mathbf{x}^*) \leq b_j \quad \text{and} \quad \lambda_j(g_j(\mathbf{x}^*) - b_j) = 0 \quad \text{for } j = 1, \ldots, m \quad (2)$$

$$x_i \geq 0 \quad \text{and} \quad \left(\frac{\partial f}{\partial x_i}(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{x})\right) x_i = 0 \quad \text{for } i = 1, \ldots, n \quad (3)$$

Optimisation Problems with Positivity Constraints

Kuhn-Tucker reformulation:

Consider the modified Lagrangian:

$$\tilde{L}(\mathbf{x};\boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j (g_j(\mathbf{x}) - b_j)$$

(Note we are not including the positivity constraints in the Lagrangian)

Kuhn-Tucker Conditions

Suppose $\mathbf{x}^*$ is a local maximum of f subject to the constraints $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$, and $\mathbf{x} \geq \mathbf{0}$. Suppose that the constraint qualification is satisfied at $\mathbf{x}^*$. Then there exists $\boldsymbol{\lambda}^*$ such that

$$D_{x_i}\tilde{L}(\mathbf{x}^*;\boldsymbol{\lambda}^*) \leq 0 \quad \text{and} \quad x_i^* D_{x_i}\tilde{L}(\mathbf{x}^*;\boldsymbol{\lambda}^*) = 0 \quad \text{for } i = 1, ..., n$$

$$\lambda_j^* \geq 0 \quad \text{and} \quad \lambda_j^*(g_j(\mathbf{x}^*) - b_j) = 0 \quad \text{for } j = 1, \ldots, m$$

You can use either formulation to solve for problems with positivity constraints.

Optimisation with Equality and Inequality Constraints

$$\max f(\mathbf{x}) \qquad \text{subject to} \quad h(\mathbf{x}) = \mathbf{a} \quad \text{and} \quad g(\mathbf{x}) \leq \mathbf{b}$$

Define the Lagrangian L as:

$$L(\mathbf{x}; \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j (h_j(\mathbf{x}) - a_j) - \sum_{j=1}^{k} \mu_j (g_j(\mathbf{x}) - b_j)$$

and similarly $D_x L(\mathbf{x}; \boldsymbol{\lambda}, \boldsymbol{\mu})$ and $D_x^2 L(\mathbf{x}; \boldsymbol{\lambda}, \boldsymbol{\mu})$.

Everything follows in the obvious way:

The constraint qualification requires that the matrix of binding constraints, $(D\mathbf{h}(\mathbf{x}^*), D\mathbf{g}_E(\mathbf{x}^*))^T$, has full rank, $m + e$.

Optimisation with Equality and Inequality Constraints

First Order Necessary Conditions

If $\mathbf{x}^*$ is a local maximum of f on S, and CQ holds, then $\mathbf{x}^*$ is a solution to:

1. the first order conditions

$$\frac{\partial L}{\partial x}(\mathbf{x}^*; \boldsymbol{\lambda}^*; \boldsymbol{\mu}^*) = 0 \qquad \text{for } i = 1, \ldots, n$$

2. the equality constraints

$$h_j(\mathbf{x}^*) = a_j \qquad \text{for } j = 1, \ldots m$$

3. the complementary slackness conditions for the inequality constraints

$$\mu_j^* \geq 0, \qquad g_j(\mathbf{x}^*) \leq b_j \quad \text{and} \quad \mu_j^*(g_j(\mathbf{x}^*) - b_j) = 0 \qquad \text{for } j = 1, \ldots, k$$

Optimisation with Equality and Inequality Constraints

Second Order Sufficient Conditions

Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ satisfy CQ and FOC.

If $\mathbf{v}^T D_x^2 L(\mathbf{x}^*; \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)\mathbf{v} < 0$ for all $\mathbf{v} \neq 0$ in $\mathbb{R}^n$ for which $Dh(\mathbf{x}^*)\mathbf{v} = 0$ and $Dg_E(\mathbf{x}^*)\mathbf{v} = 0$ then $\mathbf{x}^*$ is a strict local maximum.

The bordered Hessian is

$$\begin{pmatrix} \mathbf{0} & \mathbf{0} & Dh(\mathbf{x}^*) \\ \mathbf{0} & \mathbf{0} & Dg_E(\mathbf{x}^*) \\ Dh(\mathbf{x}^*)^T & Dg_E(\mathbf{x}^*)^T & D_x^2 L(\mathbf{x}^*; \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \end{pmatrix}$$

and you check that the signs of the determinants of the last $n - (m + e)$ principal minors alternate, ending with $(-1)^n$.

You can write the corresponding FOC and SOC for minima.

Optimisation with Equality and Inequality Constraints

To sum up:

1. Consider whether the problem has a solution (Weierstrass theorem/ function explodes in one unconstrained direction).
2. Check constraint qualification (Jacobian of (potentially) binding constraints is full rank). Save those points at which the CQ fails as potential candidates).
3. Construct the Lagrangian and solve the FOCs ($D_x L(\mathbf{x}^*, \boldsymbol{\lambda}^*) = 0$, $\mathbf{h}(\mathbf{x}^*) = \mathbf{a}$ + CS constraints to find all the candidates for interior maxima (minima).
4. If you know that your problem has a solution, it has to be one of those. Evaluate the function and find the optimal.
5. Otherwise, check the SOCs using the Bordered Hessian (with binding constraints) and reconsider the global maximum given the optimal local maximum.

An Example

Find the maximum and minimum of $x^3 + y^3 - 3x - 3y + 3xy$ subject to $x^2 \leq 4$, $y^2 \leq 4$.

1. The constraint set is compact, so a global maximum and a global minimum both exist.

2. Now consider the constraint qualification:

$$D\mathbf{g} = \begin{pmatrix} \partial g_1/\partial x & \partial g_1/\partial y \\ \partial g_2/\partial x & \partial g_2/\partial y \end{pmatrix} = \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix}$$

   - if $x^2 = 4$ and $y^2 < 4$, then $D\mathbf{g}_E = \begin{pmatrix} 2x & 0 \end{pmatrix}$ – rank 1;
   - if $x^2 < 4$ and $y^2 = 4$, then $D\mathbf{g}_E = \begin{pmatrix} 0 & 2y \end{pmatrix}$ – rank 1;
   - if $x^2 = 4$ and $y^2 = 4$, then $D\mathbf{g}_E = D\mathbf{g}$ – rank 2;

   therefore the constraint qualification cannot fail.

An Example

3 So the K-T conditions are necessary for a maximum/minimum.

We start with:

Maximise $x^3 + y^3 - 3x - 3y + 3xy$ subject to $x^2 \leq 4$, $y^2 \leq 4$

Form the Lagrangian:

$$L(x, y; \lambda_1, \lambda_2) = f(x, y) - \lambda_1(x^2 - 4) - \lambda_2(y^2 - 4)$$

$$\begin{aligned}
\text{FOC}_x: &\quad 3x^2 - 3 + 3y - 2\lambda_1 x = 0 && (1)\\
\text{FOC}_y: &\quad 3y^2 - 3 + 3x - 2\lambda_2 y = 0 && (2)\\
\text{CS}_x: &\quad \lambda_1 \geq 0, \quad x^2 \leq 4, \quad \lambda_1(x^2 - 4) = 0 && (3)\\
\text{CS}_y: &\quad \lambda_2 \geq 0, \quad y^2 \leq 4, \quad \lambda_2(y^2 - 4) = 0 && (4)
\end{aligned}$$

We now solve the K-T conditions.

An Example

(i) Neither constraint binds – we solved this problem earlier.

Strict local maximum of $\simeq \boxed{9.1}$ at $(\frac{-1-\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}) \simeq (-1.62, -1.62)$.

(ii) Both constraints bind.

(3) becomes $\lambda_1 \geq 0,\ x^2 = 4$, & (4) becomes $\lambda_2 \geq 0,\ y^2 = 4$,

- so (1) $\implies \lambda_1 = \dfrac{9+3y}{2x}$;
  the numerator is +ve, so the denominator must be +ve, and $x = +2$.
- Similarly, (2) $\implies \lambda_2 = \dfrac{9+3x}{2y} \implies y = +2$.

SOC: $n = 2, e = 2, n - e = 0$ so nothing to check.

(There are no points near $(2, 2)$ that satisfy both constraints.)

There is a constrained maximum: $f(2, 2) = \boxed{16}$.

An Example

(iii) One constraint binds.

- Say (3) becomes $\lambda_1 \geq 0,\ x^2 = 4$, & (4) becomes $\lambda_2 = 0,\ y^2 < 4$.

  As before, (1) $\implies x = +2$,

  but (2) $\implies y^2 - 1 + 2 = 0$ – no real solution.

  So, no maximum where $x^2 = 4$ & $y^2 < 4$.

- Similarly, no maximum where $y^2 = 4$ & $x^2 < 4$.

Two candidate maxima: one interior and one constrained.

Evaluate & Compare!

$\boxed{\text{Global maximum: } f(2,2) = 16}$

An Example

4 Now we look for:

$$\text{Minimise } x^3 + y^3 - 3x - 3y + 3xy \text{ subject to } x^2 \leq 4,\ y^2 \leq 4$$

Form the Lagrangian:

$$L(x, y; \lambda_1, \lambda_2) = f(x, y) + \lambda_1(x^2 - 4) + \lambda_2(y^2 - 4)$$

$$\begin{aligned}
\text{FOC}_x\text{:} &\quad 3x^2 - 3 + 3y + 2\lambda_1 x = 0 &(5)\\
\text{FOC}_y\text{:} &\quad 3y^2 - 3 + 3x + 2\lambda_2 y = 0 &(6)\\
\text{CS}_x\text{:} &\quad \lambda_1 \geq 0, \quad x^2 \leq 4, \quad \lambda_1(x^2 - 4) = 0 &(7)\\
\text{CS}_y\text{:} &\quad \lambda_2 \geq 0, \quad y^2 \leq 4, \quad \lambda_2(y^2 - 4) = 0 &(8)
\end{aligned}$$

We now solve the K-T conditions.

An Example

(i) Neither constraint binds – we solved this problem earlier.

Strict local minimum of $\simeq \boxed{-2.1}$ at $(\frac{-1+\sqrt{5}}{2}, \frac{-1+\sqrt{5}}{2}) \simeq (0.62, 0.62)$.

(ii) Both constraints bind.

$(5) \implies \lambda_1 = -\dfrac{9+3y}{2x}$, and so $x = -2$.

Also, $(6) \implies y = -2$.

SOC: nothing to check.

There is a constrained minimum: $f(-2,-2) = \boxed{8}$.

An Example

(iii) One constraint binds.

- Say (3) becomes $\lambda_1 \geq 0,\ x^2 = 4$, & (4) becomes $\lambda_2 = 0,\ y^2 < 4$.

  As before, (5) $\implies x = -2$,
  and now (6) $\implies y^2 - 1 - 2 = 0$, i.e. $y = \pm\sqrt{3}$.

$$D\mathbf{g}_E = \begin{pmatrix} 2x & 0 \end{pmatrix}; \qquad D^2 L = \begin{pmatrix} 6x + 2\lambda_1 & 3 \\ 3 & 6y + 2\lambda_2 \end{pmatrix}.$$

Bordered Hessian: $\begin{pmatrix} 0 & -4 & 0 \\ -4 & -12 + 2\lambda_1 & 3 \\ 0 & 3 & 6y \end{pmatrix}$

$n = 2, e = 1$ so check last determinant. $\{$need sign of $(-1)^e$ for a min$\}$.

$|\cdot| = -96y < 0$ when $y = +\sqrt{3}$.
So, constrained minimum: $f(-2, \sqrt{3}) \simeq \boxed{-12.4}$.

An Example

(iii) One constraint binds.

- Similarly, if (3) becomes $\lambda_1 = 0,\ x^2 < 4$, & (4) becomes $\lambda_2 \geq 0,\ y^2 = 4$.

  We obtain a constrained minimum $f(\sqrt{3}, -2) \simeq \boxed{-12.4}$.

Four candidate minima: one interior and three constrained.

Evaluate & Compare!

$$\boxed{\text{Two global minima: } f(-2, \sqrt{3}) = f(\sqrt{3}, -2) = -12.4}$$